Supplementary Information for: Carbohydrate active enzymes in *Pectobacteriaceae*: coevolving enzyme sets and host adaptation

Emma E. M. Hobbs, Tracey M. Gloster, Leighton Pritchard

2026

This document contains all supplementary information for the Manuscript: Carbohydrate active enzymes in *Pectobacteriaceae*: coevolving enzyme sets and host adaptation.

Contents

1 Methods **2**

2 Comparison of proteome size **6**

3 Comparison of the percentage of the proteome that is encapsulated in the CAZome **8**

4 Comparison of CAZome size **11**

5 Comparison of the number of CAZyme families **14**

6 Comparison of CAZy class frequencies **17**

7 Principal Component Analysis (PCA) of CAZy family frequencies **25**

8 Exploration of CAZy family frequencies **32**

9 Coinfinder output **34**

1 Methods

SI Figure 1: Method to compile a proteome dataset

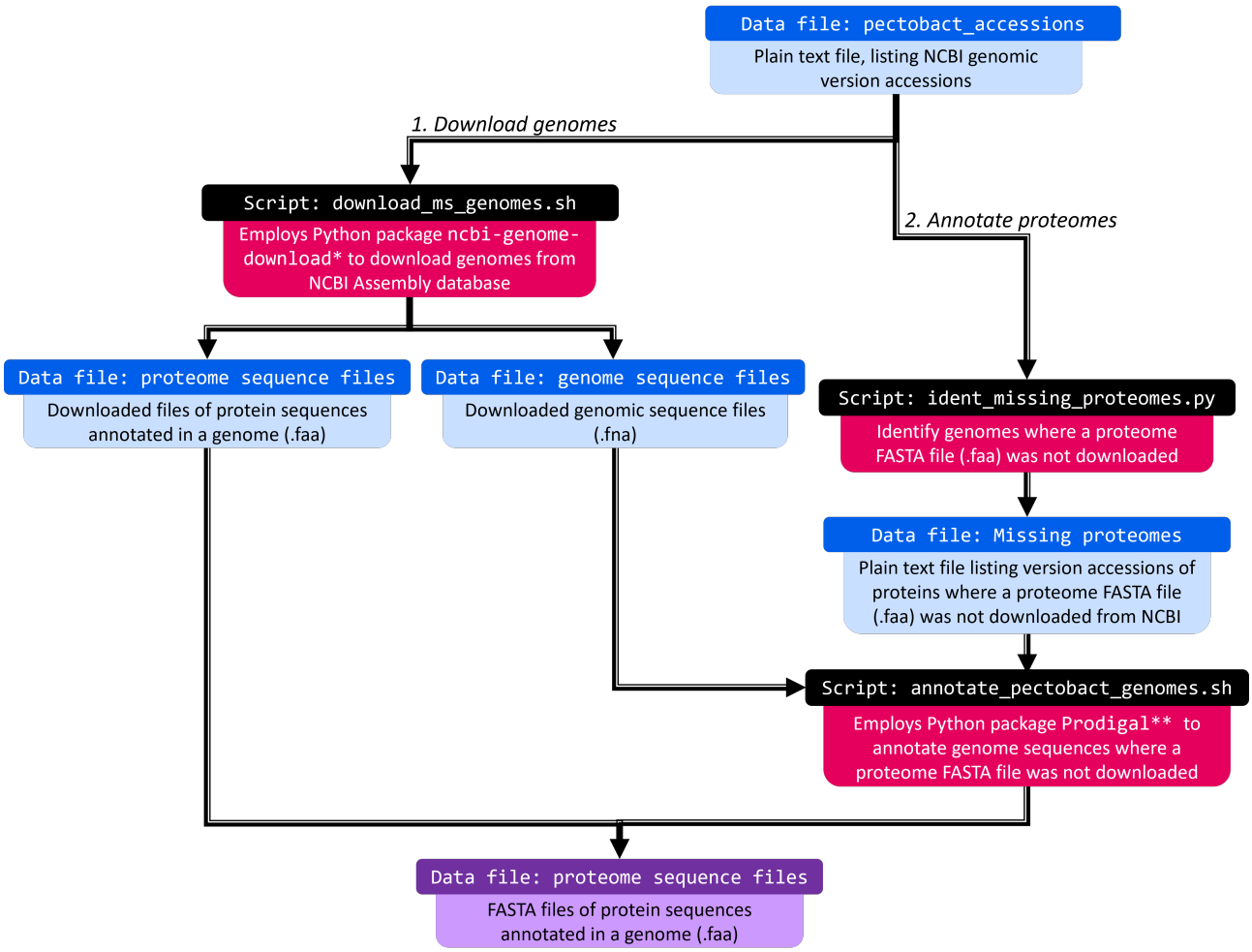


Figure 1: **Schematic summarising the method used to compile a proteome dataset for *Pectobacteriaceae*.** Bash scripts used to coordinate downloading genomes from NCBI (* using `ncbi-genome-download` [**?**]), and annotating the downloaded genomes (** using `Prodigal` [**?**]) to generate FASTA files of annotated protein sequences are shown in black. The final output is shaded in purple.

SI Figure 2: Method to compile a CAZyme dataset

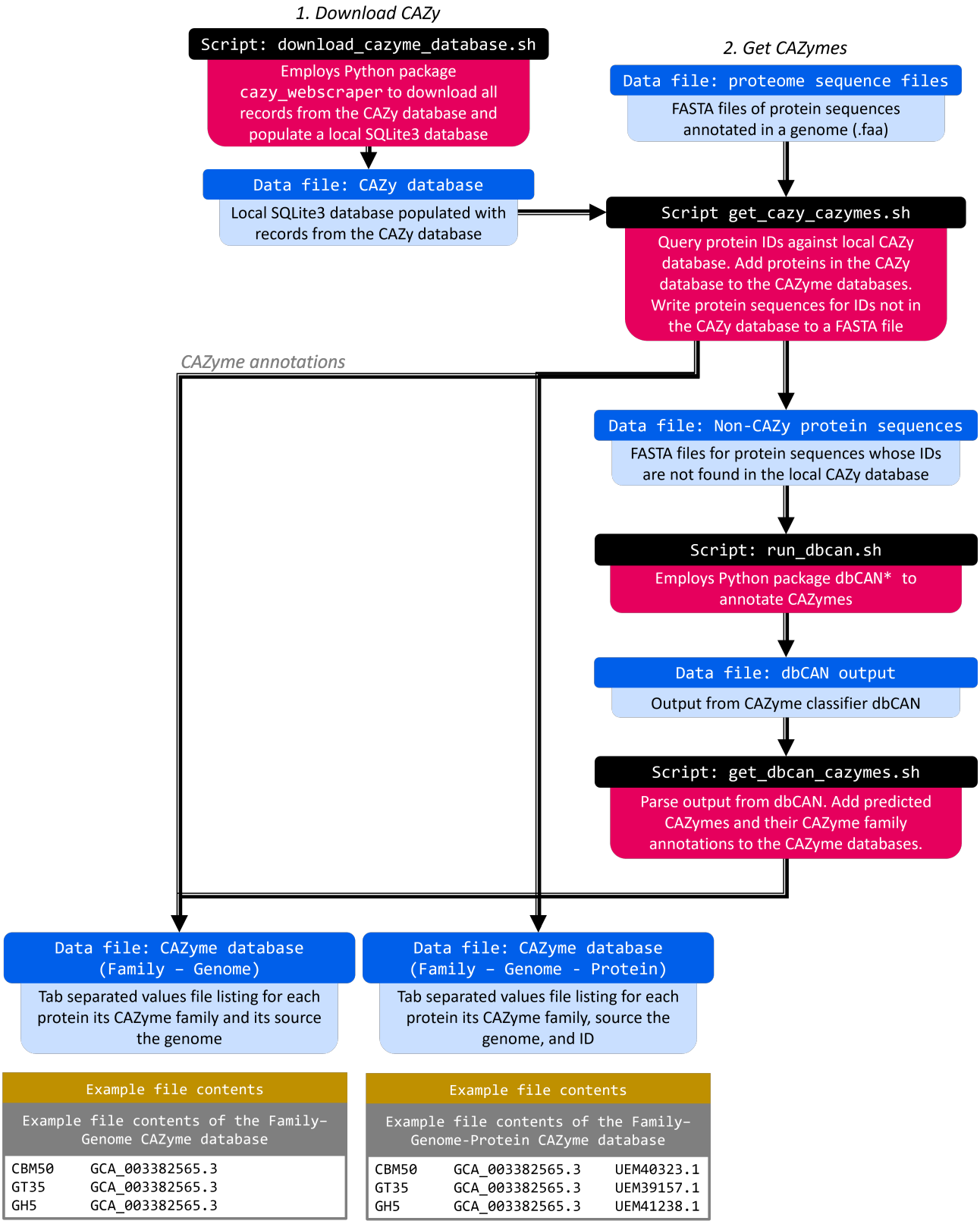


Figure 2: **Schematic summarising the method used to compile a *Pectobacteriaceae* CAZyme database**. Bash scripts (represented by black and pink boxes) used to coordinate `cazomevolve` to compile CAZyme databases listing CAZyme annotations from a local CAZy database, and predicted by dbCAN (* [**?**]). Data files generated by the pipeline are represented by blue boxes. Example output are presented in grey boxes.

SI Figure 3: Method to add taxonomic classifications

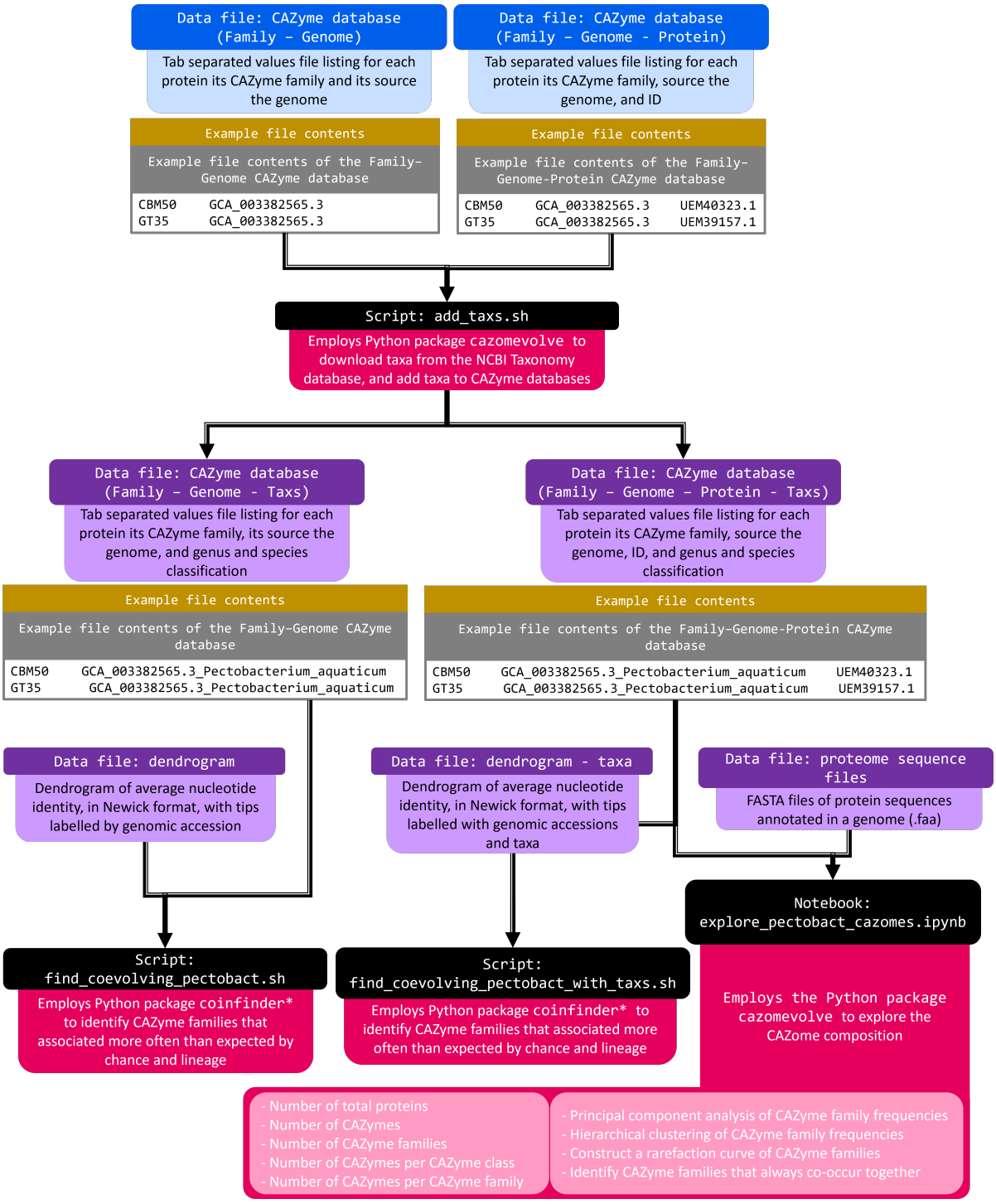


Figure 3: **Schematic summarising the methods used to add taxonomy data to the *Pectobacteriaceae* CAZyme databases** Scripts (represented by black and pink boxes) used to coordinate `cazomevolve` to download taxa (genus and species) from NCBI and add the taxa to CAZyme databases, as well as the application of the CAZyme databases within the `cazomevolve` pipeline. Operations in the pale pink boxes are performed by `cazomevolve`, configured using the notebook `explore_pectobact_cazomes.ipynb`. Blue boxes represented input datafiles. Purple boxes represent datafiles that were interrogated to compare the CAZome compositions of Pectobacteriaceae. Example content of datafiles is shown in grey boxes.

SI Figure 4: Method to build average nucleotide identity (ANI) tree

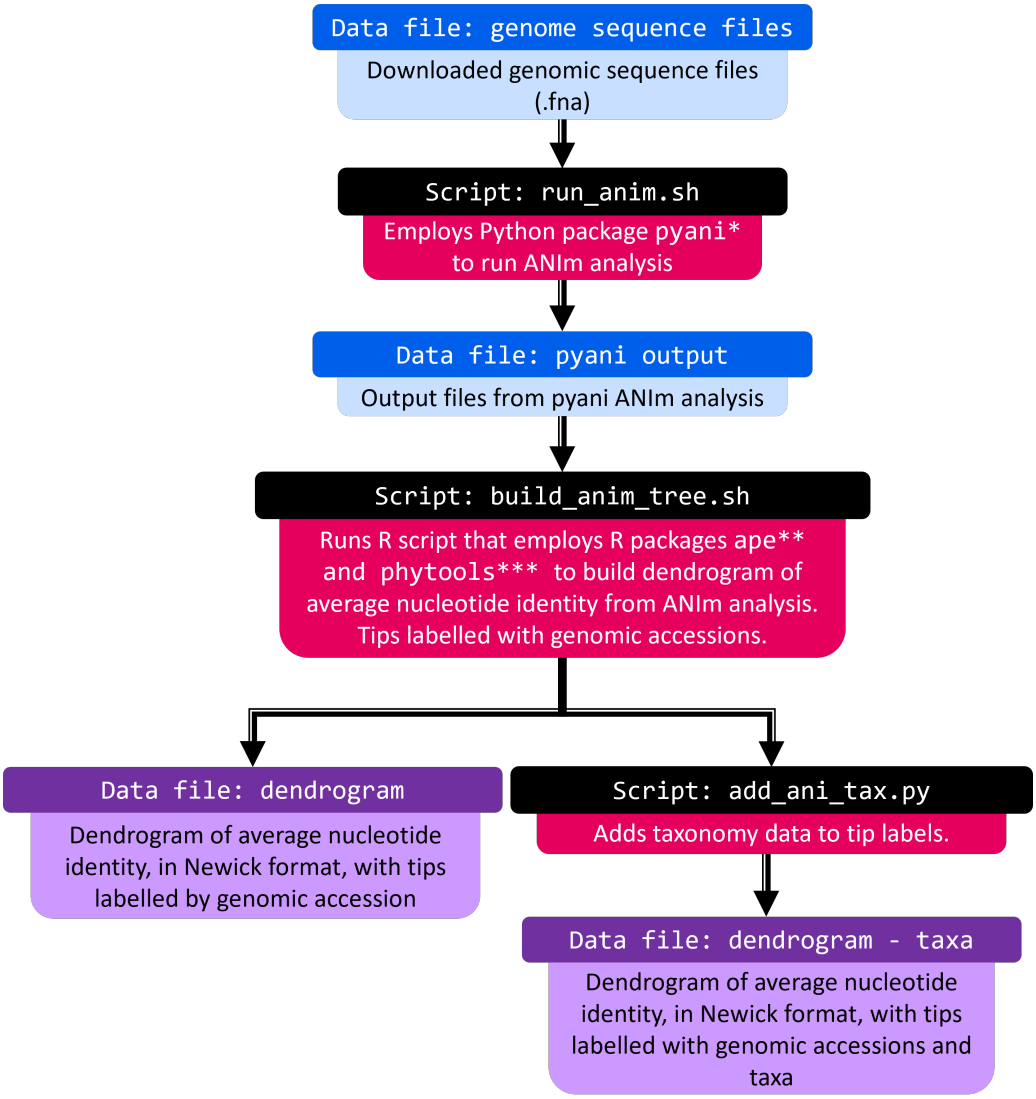


Figure 4: **Schematic summarising the method used to build a distanced-based tree for *Pectobacteriaceae* genome sequences** Scripts used to coordinate `pyani` (* [?]) to calculate the average nucleotide identity, and using the R script to build a dendrogram (** [?], *** [?]).

2 Comparison of proteome size

SI Figure 5: Boxplot overlaid by a scatter plot of the number of proteins in the proteome

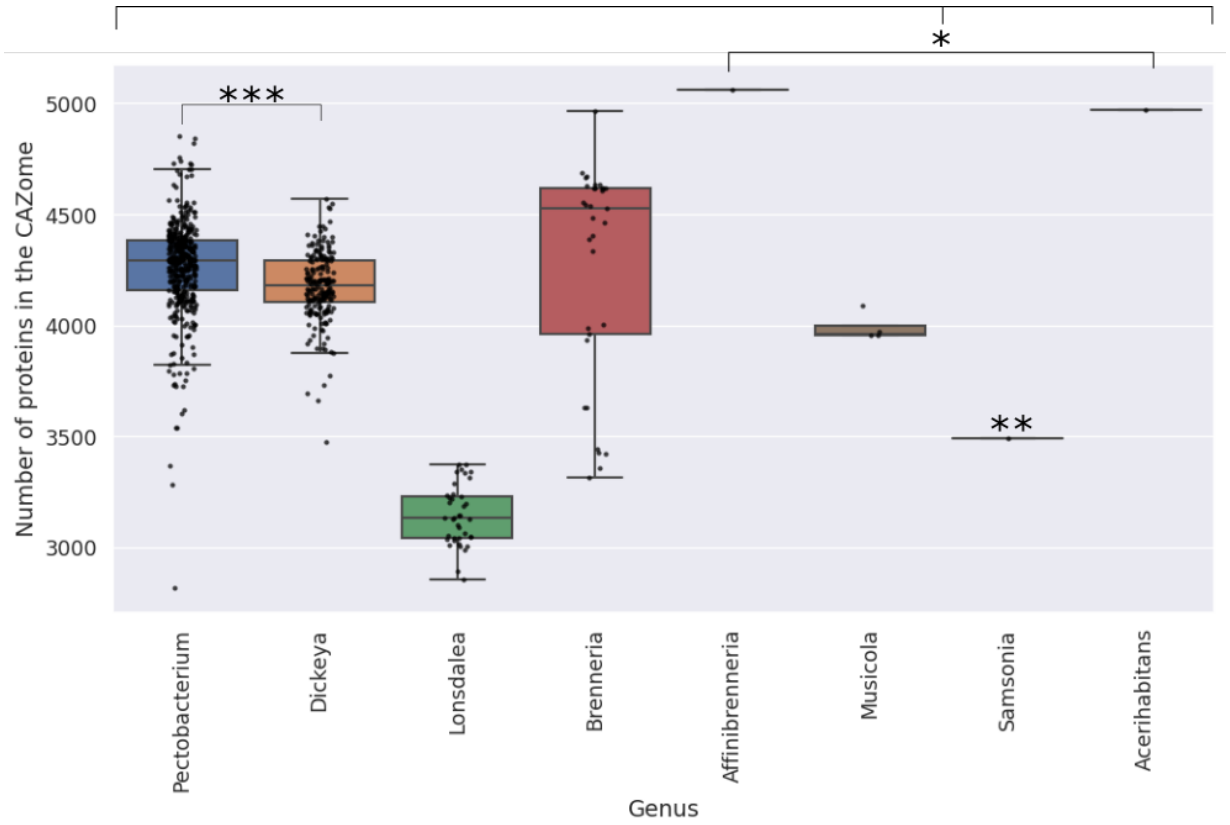


Figure 5: Number of proteins (i.e. the number of unique protein IDs) in each genome, grouped by genus which each point representing a unique genome, overlaying genomes are indicated with a darker shade.

SI Table 1: Turkey HDS post hoc test to identify genera with statistically significantly different mean proteome sizes

Output from a Turkey HDS post hoc test of the proteome sizes across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	95	1	-839.9267	1029.9267	False
Acerihabitans	Brenneria	-698.7576	0.0344	-1369.7924	-27.7227	True
Acerihabitans	Dickeya	-792.1408	0.0072	-1454.8365	-129.4451	True
Acerihabitans	Lonsdalea	-1826.7179	0	-2496.2329	-1157.203	True
Acerihabitans	Musicola	-977	0.0017	-1716.1245	-237.8755	True
Acerihabitans	Pectobacterium	-708.287	0.0262	-1370.1448	-46.4293	True
Acerihabitans	Samsonia	-1480	0	-2414.9267	-545.0733	True
Affinibrenneria	Brenneria	-793.7576	0.0083	-1464.7924	-122.7227	True
Affinibrenneria	Dickeya	-887.1408	0.0014	-1549.8365	-224.4451	True
Affinibrenneria	Lonsdalea	-1921.7179	0	-2591.2329	-1252.203	True
Affinibrenneria	Musicola	-1072	0.0003	-1811.1245	-332.8755	True
Affinibrenneria	Pectobacterium	-803.287	0.0059	-1465.1448	-141.4293	True
Affinibrenneria	Samsonia	-1575	0	-2509.9267	-640.0733	True
Brenneria	Dickeya	-93.3832	0.3004	-217.3402	30.5738	False
Brenneria	Lonsdalea	-1127.9604	0	-1284.3254	-971.5954	True
Brenneria	Musicola	-278.2424	0.2348	-628.2492	71.7644	False
Brenneria	Pectobacterium	-9.5295	1	-128.9256	109.8667	False
Brenneria	Samsonia	-781.2424	0.0101	-1452.2773	-110.2076	True
Dickeya	Lonsdalea	-1034.5772	0	-1150.0234	-919.131	True
Dickeya	Musicola	-184.8592	0.6978	-518.5995	148.881	False
Dickeya	Pectobacterium	83.8537	0.0002	27.8783	139.8292	True
Dickeya	Samsonia	-687.8592	0.0355	-1350.5549	-25.1635	True
Lonsdalea	Musicola	849.7179	0	502.634	1196.8019	True
Lonsdalea	Pectobacterium	1118.4309	0	1007.8962	1228.9657	True
Lonsdalea	Samsonia	346.7179	0.7657	-322.797	1016.2329	False
Musicola	Pectobacterium	268.713	0.2146	-63.3603	600.7863	False
Musicola	Samsonia	-503	0.4363	-1242.1245	236.1245	False
Pectobacterium	Samsonia	-771.713	0.0099	-1433.5707	-109.8552	True

3 Comparison of the percentage of the proteome that is encapsulated in the CAZome

subsection*SI Figure 6: Boxplot overlaid by a scatter plot of the percentage of the proteome encapsulated by the CAZome

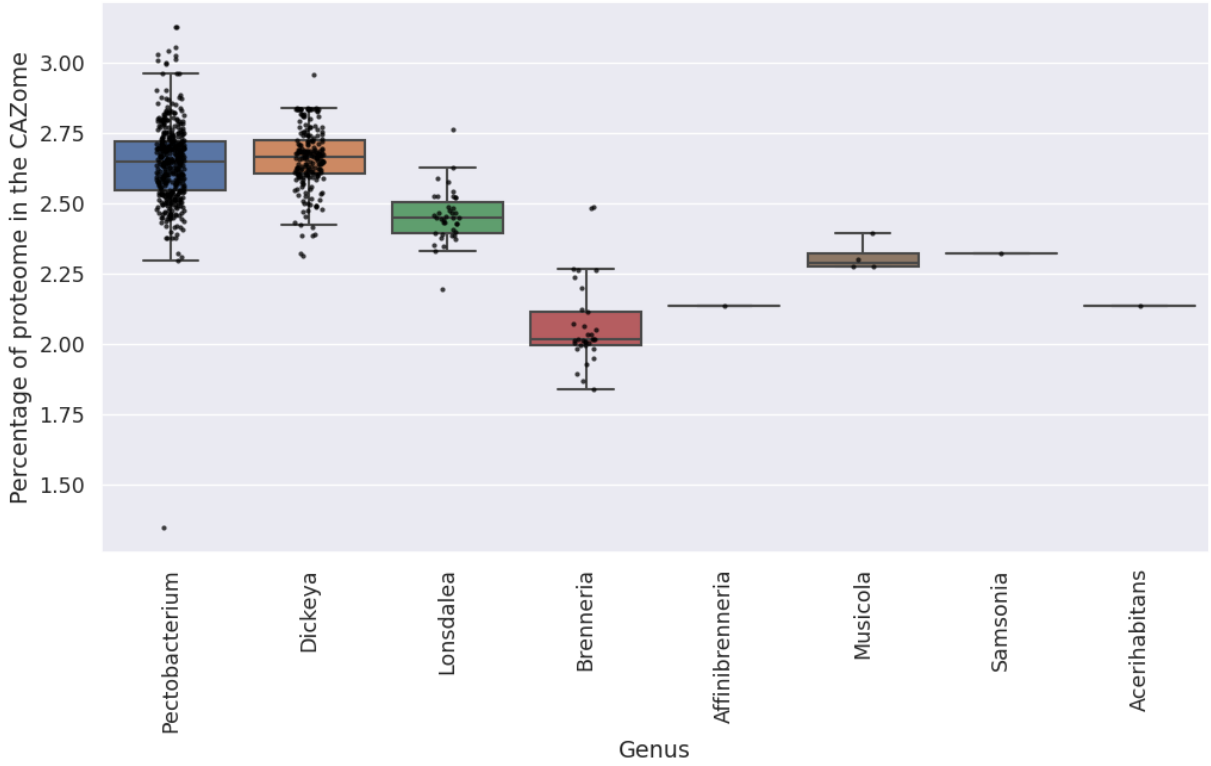


Figure 6: The percentage of the proteome encapsulated by the CAZome in each genome, grouped by genus which each point representing a unique genome, overlaying genomes are indicated with a darker shade.

SI Table 2: Turkey HDS post hoc test to identify statistically significant differences in mean percentage of the proteome that is encapsulated by the CAZome in soft and hard plant tissue targeting genomes

Output from a Turkey HDS post hoc test of the percentage of the proteome that is encapsulated by the CAZome across soft plant tissue targeting genera (*Pectobacterium* and *Dickeya*) and hard plant tissue targeting genera (*Acerhabitans*, *Affinibrenneria*, *Brenneria*, *Lonsdalea*, and *Samsonia*). *Musicola* is labelled as 'unknown' owing to its preference of soft and hard plant tissues not having yet been established in the literature. A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Hard	Soft	0.3753	0	0.3331	0.4175	True
Hard	Unknown	0.0371	0.8757	-0.1404	0.2145	False
Soft	Unknown	-0.3382	0	-0.5117	-0.1648	True

SI Table 3: Turkey HDS post hoc test to identify genera with statistically significantly different mean percentage of the proteome encapsulated in the CAZome

Output from a Turkey HDS post hoc test of the percentage of the proteome represented by the CAZome across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	-0.0005	1	-0.5762	0.5751	False
Acerihabitans	Brenneria	-0.0662	0.9997	-0.4793	0.347	False
Acerihabitans	Dickeya	0.5273	0.0024	0.1192	0.9353	True
Acerihabitans	Lonsdalea	0.3215	0.2572	-0.0907	0.7337	False
Acerihabitans	Musicola	0.1777	0.9356	-0.2774	0.6327	False
Acerihabitans	Pectobacterium	0.5105	0.0038	0.103	0.918	True
Acerihabitans	Samsonia	0.1884	0.9752	-0.3873	0.764	False
Affinibrenneria	Brenneria	-0.0656	0.9997	-0.4788	0.3475	False
Affinibrenneria	Dickeya	0.5278	0.0023	0.1198	0.9358	True
Affinibrenneria	Lonsdalea	0.322	0.2553	-0.0902	0.7343	False
Affinibrenneria	Musicola	0.1782	0.9347	-0.2769	0.6333	False
Affinibrenneria	Pectobacterium	0.511	0.0037	0.1035	0.9185	True
Affinibrenneria	Samsonia	0.1889	0.9748	-0.3868	0.7645	False
Brenneria	Dickeya	0.5934	0	0.5171	0.6697	True
Brenneria	Lonsdalea	0.3877	0	0.2914	0.484	True
Brenneria	Musicola	0.2438	0.0142	0.0283	0.4593	True
Brenneria	Pectobacterium	0.5767	0	0.5031	0.6502	True
Brenneria	Samsonia	0.2545	0.5702	-0.1586	0.6677	False
Dickeya	Lonsdalea	-0.2057	0	-0.2768	-0.1347	True
Dickeya	Musicola	-0.3496	0	-0.5551	-0.1441	True
Dickeya	Pectobacterium	-0.0168	0.8188	-0.0512	0.0177	False
Dickeya	Samsonia	-0.3389	0.1866	-0.7469	0.0691	False
Lonsdalea	Musicola	-0.1439	0.4512	-0.3576	0.0698	False
Lonsdalea	Pectobacterium	0.189	0	0.1209	0.257	True
Lonsdalea	Samsonia	-0.1332	0.9769	-0.5454	0.2791	False
Musicola	Pectobacterium	0.3328	0	0.1284	0.5373	True
Musicola	Samsonia	0.0107	1	-0.4444	0.4658	False
Pectobacterium	Samsonia	-0.3221	0.2413	-0.7296	0.0854	False

4 Comparison of CAZome size

SI Figure 7: Boxplot overlaid by a scatter plot of the size of the CAZome

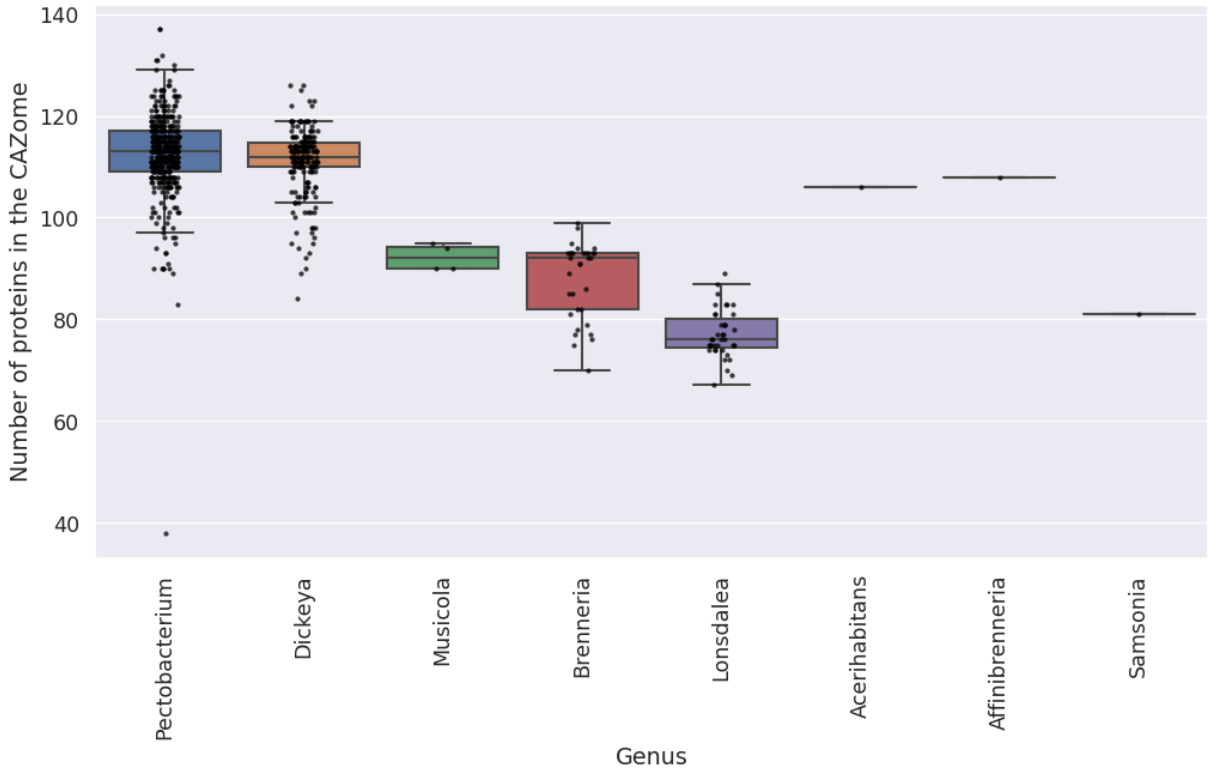


Figure 7: The number of CAZymes in the CAZome (i.e. the unique protein IDs in the CAZome) in each genome, grouped by genus which each point representing a unique genome, overlaying genomes are indicated with a darker shade.

SI Table 4: Turkey HDS post hoc test to identify statistically significant differences in mean CAZome size in soft and hard plant tissue targeting genomes

Output from a Turkey HDS post hoc test of the CAZome size (number of unique protein IDs in the CAZome) across soft plant tissue targeting genera (*Pectobacterium* and *Dickeya*) and hard plant tissue targeting genera (*Acerhabitans*, *Affinibrenneria*, *Brenneria*, *Lonsdalea*, and *Samsonia*). *Musicola* is labelled as 'unknown' owing to its preference of soft and hard plant tissues not having yet been established in the literature. A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Hard	Soft	29.4861	0	27.2598	31.7125	True
Hard	Unknown	9.57	0.0437	0.2108	18.9292	True
Soft	Unknown	-19.91661	0	-29.0639	-10.7684	True

SI Table 5: Turkey HDS post hoc test to identify genera with statistically significantly different mean CAZome sizes

Output from a Turkey HDS post hoc test of the CAZome sizes (i.e. number of unique protein IDs) across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	2	1	-30.0935	34.0935	False
Acerihabitans	Brenneria	-18.2121	0.2411	-41.2469	4.8227	False
Acerihabitans	Dickeya	5.1602	0.9973	-17.5883	27.9087	False
Acerihabitans	Lonsdalea	-28.8462	0.0037	-51.8288	-5.8635	True
Acerihabitans	Musicola	-13.75	0.7211	-39.1221	11.6221	False
Acerihabitans	Pectobacterium	6.6458	0.987	-16.0739	29.3656	False
Acerihabitans	Samsonia	-25	0.2588	-57.0935	7.0935	False
Affinibrenneria	Brenneria	-20.2121	0.1344	-43.2469	2.8227	False
Affinibrenneria	Dickeya	3.1602	0.9999	-19.5883	25.9087	False
Affinibrenneria	Lonsdalea	-30.8462	0.0013	-53.8288	-7.8635	True
Affinibrenneria	Musicola	-15.75	0.5603	-41.1221	9.6221	False
Affinibrenneria	Pectobacterium	4.6458	0.9986	-18.0739	27.3656	False
Affinibrenneria	Samsonia	-27	0.1735	-59.0935	5.0935	False
Brenneria	Dickeya	23.3723	0	19.1172	27.6274	True
Brenneria	Lonsdalea	-10.634	0	-16.0016	-5.2664	True
Brenneria	Musicola	4.4621	0.9504	-7.5527	16.4769	False
Brenneria	Pectobacterium	24.858	0	20.7594	28.9565	True
Brenneria	Samsonia	-6.7879	0.9864	-29.8227	16.2469	False
Dickeya	Lonsdalea	-34.0063	0	-37.9693	-30.0434	True
Dickeya	Musicola	-18.9102	0	-30.3666	-7.4538	True
Dickeya	Pectobacterium	1.4856	0.2679	-0.4358	3.4071	False
Dickeya	Samsonia	-30.1602	0.0016	-52.9087	-7.4117	True
Lonsdalea	Musicola	15.0962	0.0032	3.1817	27.0106	True
Lonsdalea	Pectobacterium	35.492	0	31.6976	39.2863	True
Lonsdalea	Samsonia	3.8462	0.9996	-19.1365	26.8288	False
Musicola	Pectobacterium	20.3958	0	8.9967	31.795	True
Musicola	Samsonia	-11.25	0.88	-36.6221	14.1221	False
Pectobacterium	Samsonia	-31.6458	0.0007	-54.3656	-8.9261	True

5 Comparison of the number of CAZyme families

SI Figure 8: Boxplot overlaid by a scatter plot of the number of CAZy families in each genome

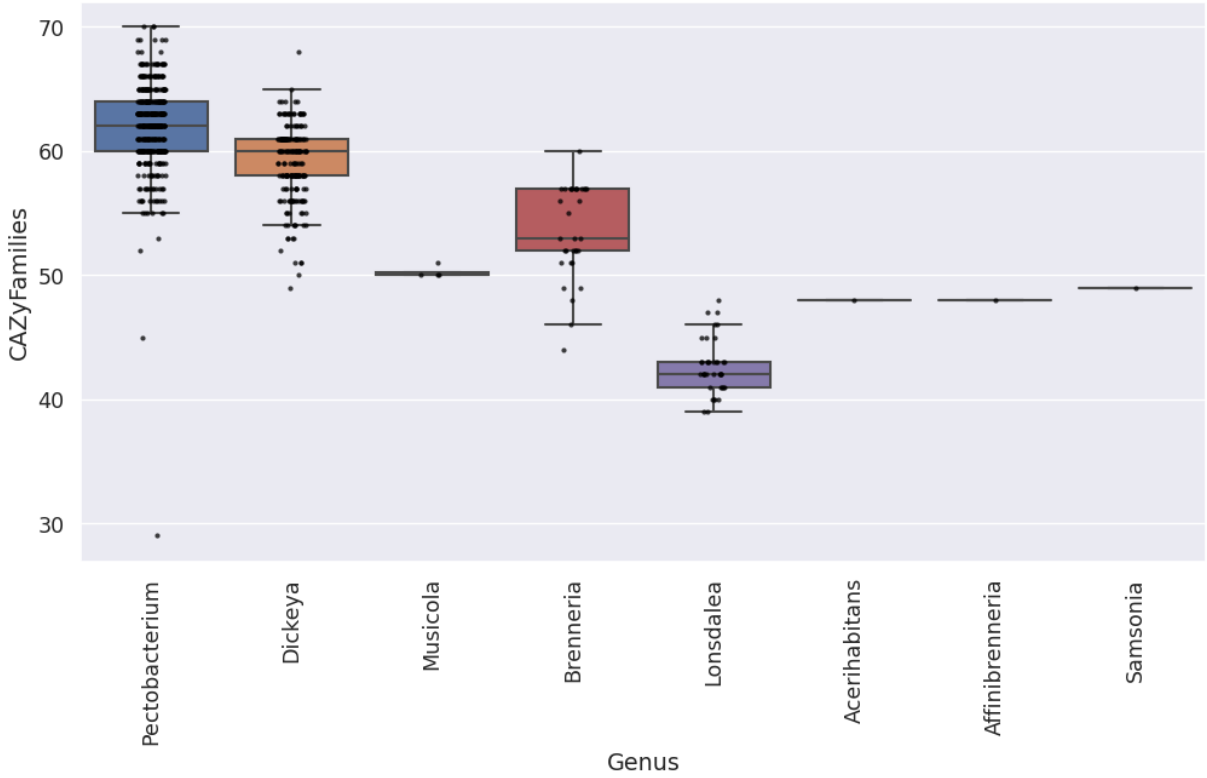


Figure 8: The number of CAZyme families in the CAZome of each genome, grouped by genus which each point representing a unique genome, overlaying genomes are indicated with a darker shade.

SI Table 6: Turkey HDS post hoc test to identify statistically significant differences in mean number of CAZy families in the CAZome in soft and hard plant tissue targeting genomes

Output from a Turkey HDS post hoc test of the mean number of CAZyme families in the CAZome across soft plant tissue targeting genera (*Pectobacterium* and *Dickeya*) and hard plant tissue targeting genera (*Acerhabitans*, *Affinibrenneria*, *Brenneria*, *Lonsdalea*, and *Samsonia*). *Musicola* is labelled as 'unknown' owing to its preference of soft and hard plant tissues not having yet been established in the literature. A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Hard	Soft	13.3936	0	12.593	14.528	True
Hard	Unknown	2.5433	0.4226	-2.2253	7.312	False
Soft	Unknown	-10.8503	0	-15.5112	-6.1894	True

SI Table 7: Turkey HDS post hoc test to identify genera with statistically significantly different mean number of CAZyme families

Output from a Turkey HDS post hoc test of the number of CAZy families in each CAZome, across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group1	**Group2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	0	1	-14.1792	14.1792	False
Acerihabitans	Brenneria	5.6667	0.6921	-4.5103	15.8437	False
Acerihabitans	Dickeya	11.0485	0.0197	0.998	21.0991	True
Acerihabitans	Lonsdalea	-5.3846	0.743	-15.5386	4.7693	False
Acerihabitans	Musicola	2.25	0.9987	-8.9597	13.4597	False
Acerihabitans	Pectobacterium	14.0787	0.0006	4.0409	24.1165	True
Acerihabitans	Samsonia	1	1	-13.1792	15.1792	False
Affinibrenneria	Brenneria	5.6667	0.6921	-4.5103	15.8437	False
Affinibrenneria	Dickeya	11.0485	0.0197	0.998	21.0991	True
Affinibrenneria	Lonsdalea	-5.3846	0.743	-15.5386	4.7693	False
Affinibrenneria	Musicola	2.25	0.9987	-8.9597	13.4597	False
Affinibrenneria	Pectobacterium	14.0787	0.0006	4.0409	24.1165	True
Affinibrenneria	Samsonia	1	1	-13.1792	15.1792	False
Brenneria	Dickeya	5.3819	0	3.5019	7.2618	True
Brenneria	Lonsdalea	-11.0513	0	-13.4227	-8.6798	True
Brenneria	Musicola	-3.4167	0.5122	-8.7249	1.8916	False
Brenneria	Pectobacterium	8.412	0	6.6013	10.2228	True
Brenneria	Samsonia	-4.6667	0.86	-14.8437	5.5103	False
Dickeya	Lonsdalea	-16.4332	0	-18.184	-14.6823	True
Dickeya	Musicola	-8.7985	0	-13.8601	-3.737	True
Dickeya	Pectobacterium	3.0302	0	2.1812	3.8791	True
Dickeya	Samsonia	-10.0485	0.0501	-20.0991	0.002	False
Lonsdalea	Musicola	7.6346	0.0003	2.3707	12.8985	True
Lonsdalea	Pectobacterium	19.4633	0	17.7869	21.1397	True
Lonsdalea	Samsonia	6.3846	0.5434	-3.7693	16.5386	False
Musicola	Pectobacterium	11.8287	0	6.7924	16.865	True
Musicola	Samsonia	-1.25	1	-12.4597	9.9597	False
Pectobacterium	Samsonia	-13.0787	0.0021	-23.1165	-3.0409	True

6 Comparison of CAZy class frequencies

SI Table 8: Two-way ANOVA of CAZy class frequencies

Output from a two-way ANOVA (performed using the Python StatsModels package, see Methods) testing for statistically significantly differences in the mean frequencies between the CAZy classes and *Pectobacteriaceae* genera. A PR(≥F) value (the p-value) of 0.05 was used as the cut-off.

Category	**Sum of squares**	**Degrees of freedom**	**F value**	**PR(>F)**
Genus	1.37E4	7	137.43	4.38E-183
CAZy Class	1.11E6	5	15606.45	0
Genus:CAZy Class	2.4E4	35	48.05	1.70E-276
Residuals	6.07e+04	4254	NA	NA

SI Table 9: Turkey HSD test of Glycoside Hydrolase (GH) CAZy class frequencies

Output from a Turkey HDS post hoc test of the number of CAZymes in the Glycoside Hydrolase (GH) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	**Group 2**	**MeanDiff**	**Adjusted P-value**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	8	0.9884	-19.8655	35.8655	False
Acerihabitans	Brenneria	-8.5152	0.901	-28.5153	11.485	False
Acerihabitans	Dickeya	-7.6699	0.9374	-27.4215	12.0817	False
Acerihabitans	Lonsdalea	-20.1538	0.0458	-40.1087	-0.199	True
Acerihabitans	Musicola	-14	0.5293	-36.0296	8.0296	False
Acerihabitans	Pectobacterium	-0.1551	1	-19.8818	19.5716	False
Acerihabitans	Samsonia	-17	0.5827	-44.8655	10.8655	False
Affinibrenneria	Brenneria	-16.5152	0.1927	-36.5153	3.485	False
Affinibrenneria	Dickeya	-15.6699	0.2371	-35.4215	4.0817	False
Affinibrenneria	Lonsdalea	-28.1538	0.0005	-48.1087	-8.199	True
Affinibrenneria	Musicola	-22	0.0506	-44.0296	0.0296	False
Affinibrenneria	Pectobacterium	-8.1551	0.9142	-27.8818	11.5716	False
Affinibrenneria	Samsonia	-25	0.1161	-52.8655	2.8655	False
Brenneria	Dickeya	0.8452	0.9971	-2.8493	4.5398	False
Brenneria	Lonsdalea	-11.6387	0	-16.2992	-6.9782	True
Brenneria	Musicola	-5.4848	0.7514	-15.9168	4.9471	False
Brenneria	Pectobacterium	8.3601	0	4.8015	11.9187	True
Brenneria	Samsonia	-8.4848	0.9027	-28.485	11.5153	False
Dickeya	Lonsdalea	-12.4839	0	-15.9248	-9.0431	True
Dickeya	Musicola	-6.3301	0.5274	-16.2772	3.617	False
Dickeya	Pectobacterium	7.5148	0	5.8465	9.1832	True
Dickeya	Samsonia	-9.3301	0.8403	-29.0817	10.4215	False
Lonsdalea	Musicola	6.1538	0.6146	-4.191	16.4987	False
Lonsdalea	Pectobacterium	19.9988	0	16.7043	23.2932	True
Lonsdalea	Samsonia	3.1538	0.9997	-16.801	23.1087	False
Musicola	Pectobacterium	13.8449	0.0006	3.9475	23.7423	True
Musicola	Samsonia	-3	0.9999	-25.0296	19.0296	False
Pectobacterium	Samsonia	-16.8449	0.159	-36.5716	2.8818	False

SI Table 10: Turkey HSD test of Glycosyltransferase (GT) CAZy class frequencies

Output from a Turkey HDS post hoc test of the number of CAZymes in the Glycosyltransferase (GT) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	**Group 2**	**MeanDiff**	**Adjusted P-value**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	-9	0.9555	-33.7332	15.7332	False
Acerihabitans	Brenneria	-11.6667	0.484	-29.4187	6.0853	False
Acerihabitans	Dickeya	-6.0437	0.9668	-23.5751	11.4877	False
Acerihabitans	Lonsdalea	-11.5385	0.4959	-29.2502	6.1733	False
Acerihabitans	Musicola	-13	0.4682	-32.5533	6.5533	False
Acerihabitans	Pectobacterium	-11.7361	0.4571	-29.2453	5.7731	False
Acerihabitans	Samsonia	-19	0.2758	-43.7332	5.7332	False
Affinibrenneria	Brenneria	-2.6667	0.9998	-20.4187	15.0853	False
Affinibrenneria	Dickeya	2.9563	0.9996	-14.5751	20.4877	False
Affinibrenneria	Lonsdalea	-2.5385	0.9999	-20.2502	15.1733	False
Affinibrenneria	Musicola	-4	0.9986	-23.5533	15.5533	False
Affinibrenneria	Pectobacterium	-2.7361	0.9998	-20.2453	14.7731	False
Affinibrenneria	Samsonia	-10	0.9232	-34.7332	14.7332	False
Brenneria	Dickeya	5.623	0	2.3437	8.9022	True
Brenneria	Lonsdalea	0.1282	1	-4.0084	4.2648	False
Brenneria	Musicola	-1.3333	0.9999	-10.5926	7.926	False
Brenneria	Pectobacterium	-0.0694	1	-3.228	3.0891	False
Brenneria	Samsonia	-7.3333	0.9145	-25.0853	10.4187	False
Dickeya	Lonsdalea	-5.4948	0	-8.5489	-2.4407	True
Dickeya	Musicola	-6.9563	0.2452	-15.7853	1.8727	False
Dickeya	Pectobacterium	-5.6924	0	-7.1732	-4.2116	True
Dickeya	Samsonia	-12.9563	0.3252	-30.4877	4.5751	False
Lonsdalea	Musicola	-1.4615	0.9997	-10.6435	7.7204	False
Lonsdalea	Pectobacterium	-0.1976	1	-3.1218	2.7265	False
Lonsdalea	Samsonia	-7.4615	0.9059	-25.1733	10.2502	False
Musicola	Pectobacterium	1.2639	0.9999	-7.521	10.0488	False
Musicola	Samsonia	-6	0.9828	-25.5533	13.5533	False
Pectobacterium	Samsonia	-7.2639	0.9127	-24.7731	10.2453	False

SI Table 11: Turkey HSD test of Carbohydrate Esterase (CE) CAZy class frequencies

Output from a Turkey HDS post hoc test of the number of CAZymes in the Carbohydrate Esterase (CE) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	**Group 2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	2	0.8959	-2.6483	6.6483	False
Acerihabitans	Brenneria	-0.697	0.9984	-4.0332	2.6393	False
Acerihabitans	Dickeya	2.2767	0.4155	-1.0181	5.5715	False
Acerihabitans	Lonsdalea	-1.8462	0.6964	-5.1749	1.4826	False
Acerihabitans	Musicola	1	0.9916	-2.6748	4.6748	False
Acerihabitans	Pectobacterium	2.2176	0.4497	-1.0731	5.5082	False
Acerihabitans	Samsonia	3	0.5085	-1.6483	7.6483	False
Affinibrenneria	Brenneria	-2.697	0.2157	-6.0332	0.6393	False
Affinibrenneria	Dickeya	0.2767	1	-3.0181	3.5715	False
Affinibrenneria	Lonsdalea	-3.8462	0.0111	-7.1749	-0.5174	True
Affinibrenneria	Musicola	-1	0.9916	-4.6748	2.6748	False
Affinibrenneria	Pectobacterium	0.2176	1	-3.0731	3.5082	False
Affinibrenneria	Samsonia	1	0.998	-3.6483	5.6483	False
Brenneria	Dickeya	2.9737	0	2.3574	3.59	True
Brenneria	Lonsdalea	-1.1492	0.0002	-1.9266	-0.3718	True
Brenneria	Musicola	1.697	0.062	-0.0432	3.4371	False
Brenneria	Pectobacterium	2.9146	0	2.3209	3.5082	True
Brenneria	Samsonia	3.697	0.018	0.3607	7.0332	True
Dickeya	Lonsdalea	-4.1229	0	-4.6968	-3.5489	True
Dickeya	Musicola	-1.2767	0.2739	-2.936	0.3826	False
Dickeya	Pectobacterium	-0.0591	0.9982	-0.3374	0.2192	False
Dickeya	Samsonia	0.7233	0.9978	-2.5715	4.0181	False
Lonsdalea	Musicola	2.8462	0	1.1205	4.5718	True
Lonsdalea	Pectobacterium	4.0637	0	3.5142	4.6133	True
Lonsdalea	Samsonia	4.8462	0.0003	1.5174	8.1749	True
Musicola	Pectobacterium	1.2176	0.3279	-0.4334	2.8686	False
Musicola	Samsonia	2	0.7166	-1.6748	5.6748	False
Pectobacterium	Samsonia	0.7824	0.9963	-2.5082	4.0731	False

SI Table 12: Turkey HSD test of Carbohydrate Binding Module (CBM) CAZy class frequencies

Output from a Turkey HDS post hoc test of the number of CAZymes in the Carbohydrate Binding Module (CBM) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	Group 2	MeanDiff	P-Adj	Lower	Upper	Reject NH
Acerihabitans	Affinibrenneria	-1	0.9999	-8.6335	6.6335	False
Acerihabitans	Brenneria	-1.6364	0.9853	-7.1152	3.8425	False
Acerihabitans	Dickeya	1.4272	0.993	-3.9836	6.838	False
Acerihabitans	Lonsdalea	-2.2564	0.9148	-7.7229	3.2101	False
Acerihabitans	Musicola	0	1	-6.0348	6.0348	False
Acerihabitans	Pectobacterium	3.1204	0.6506	-2.2836	8.5243	False
Acerihabitans	Samsonia	-1	0.9999	-8.6335	6.6335	False
Affinibrenneria	Brenneria	-0.6364	1	-6.1152	4.8425	False
Affinibrenneria	Dickeya	2.4272	0.8734	-2.9836	7.838	False
Affinibrenneria	Lonsdalea	-1.2564	0.997	-6.7229	4.2101	False
Affinibrenneria	Musicola	1	0.9996	-5.0348	7.0348	False
Affinibrenneria	Pectobacterium	4.1204	0.2852	-1.2836	9.5243	False
Affinibrenneria	Samsonia	0	1	-7.6335	7.6335	False
Brenneria	Dickeya	3.0635	0	2.0515	4.0756	True
Brenneria	Lonsdalea	-0.62	0.8199	-1.8967	0.6566	False
Brenneria	Musicola	1.6364	0.6604	-1.2214	4.4941	False
Brenneria	Pectobacterium	4.7567	0	3.7819	5.7316	True
Brenneria	Samsonia	0.6364	1	-4.8425	6.1152	False
Dickeya	Lonsdalea	-3.6836	0	-4.6262	-2.741	True
Dickeya	Musicola	-1.4272	0.755	-4.1521	1.2977	False
Dickeya	Pectobacterium	1.6932	0	1.2362	2.1502	True
Dickeya	Samsonia	-2.4272	0.8734	-7.838	2.9836	False
Lonsdalea	Musicola	2.2564	0.2329	-0.5775	5.0903	False
Lonsdalea	Pectobacterium	5.3768	0	4.4743	6.2793	True
Lonsdalea	Samsonia	1.2564	0.997	-4.2101	6.7229	False
Musicola	Pectobacterium	3.1204	0.0116	0.4091	5.8317	True
Musicola	Samsonia	-1	0.9996	-7.0348	5.0348	False
Pectobacterium	Samsonia	-4.1204	0.2852	-9.5243	1.2836	False

SI Table 13: Turkey HSD test of Carbohydrate Binding Module (CBM) CAZy class frequencies between soft and hard plant tissue targeting genera

Output from a Turkey HDS post hoc test of the number of CAZymes in the Carbohydrate Binding Module (CBM) CAZy class across soft and hard plant tissue targeting genera in *Pectobacteriaceae* and *Musicola*, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	Group 2	MeanDiff	P-Adj	Lower	Upper	Reject NH
Hard	Soft	4.4937	0	3.9416	5.0457	True
Hard	Unknown	1.92	0.1275	-0.4008	4.2408	False
Soft	Unknown	-2.5737	0.0215	-4.842	-0.3053	True

SI Table 14: Turkey HSD test of Auxiliary Activity (AA) CAZy class frequencies

Output from a Turkey HDS post hoc test of the number of CAZymes in the Auxiliary Activity (AA) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	Group 2	MeanDiff	P-Adj	Lower	Upper	Reject NH
Acerihabitans	Affinibrenneria	0	1	-1.7994	1.7994	False
Acerihabitans	Brenneria	0.2424	0.9992	-1.0491	1.5339	False
Acerihabitans	Dickeya	0.4126	0.9767	-0.8628	1.6881	False
Acerihabitans	Lonsdalea	0	1	-1.2886	1.2886	False
Acerihabitans	Musicola	0	1	-1.4225	1.4225	False
Acerihabitans	Pectobacterium	0.8843	0.4092	-0.3896	2.1581	False
Acerihabitans	Samsonia	1	0.6942	-0.7994	2.7994	False
Affinibrenneria	Brenneria	0.2424	0.9992	-1.0491	1.5339	False
Affinibrenneria	Dickeya	0.4126	0.9767	-0.8628	1.6881	False
Affinibrenneria	Lonsdalea	0	1	-1.2886	1.2886	False
Affinibrenneria	Musicola	0	1	-1.4225	1.4225	False
Affinibrenneria	Pectobacterium	0.8843	0.4092	-0.3896	2.1581	False
Affinibrenneria	Samsonia	1	0.6942	-0.7994	2.7994	False
Brenneria	Dickeya	0.1702	0.3721	-0.0684	0.4088	False
Brenneria	Lonsdalea	-0.2424	0.2197	-0.5434	0.0585	False
Brenneria	Musicola	-0.2424	0.958	-0.9161	0.4312	False
Brenneria	Pectobacterium	0.6418	0	0.412	0.8716	True
Brenneria	Samsonia	0.7576	0.6317	-0.5339	2.0491	False
Dickeya	Lonsdalea	-0.4126	0	-0.6348	-0.1904	True
Dickeya	Musicola	-0.4126	0.5148	-1.055	0.2297	False
Dickeya	Pectobacterium	0.4716	0	0.3639	0.5794	True
Dickeya	Samsonia	0.5874	0.8573	-0.6881	1.8628	False
Lonsdalea	Musicola	0	1	-0.668	0.668	False
Lonsdalea	Pectobacterium	0.8843	0	0.6715	1.097	True
Lonsdalea	Samsonia	1	0.2634	-0.2886	2.2886	False
Musicola	Pectobacterium	0.8843	0.0008	0.2451	1.5234	True
Musicola	Samsonia	1	0.392	-0.4225	2.4225	False
Pectobacterium	Samsonia	0.1157	1	-1.1581	1.3896	False

SI Table 15: Turkey HSD test of Auxiliary Activity (AA) CAZy class frequencies between soft and hard plant tissue targeting genera

Output from a Turkey HDS post hoc test of the number of CAZymes in the Auxiliary Activity (AA) CAZy class across soft and hard plant tissue targeting genera in *Pectobacteriaceae* and *Musicola*, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	Group 2	MeanDiff	P-Adj	Lower	Upper	Reject NH
Hard	Soft	0.612	0	0.4775	0.7464	True
Hard	Unknown	-0.12	0.872	-0.6853	0.4453	False
Soft	Unknown	-0.732	0.0055	-1.2845	-0.1794	True

SI Table 16: Turkey HSD test of Polysaccharide Lyase (PL) CAZy class frequencies between soft and hard plant tissue targeting genera

Output from a Turkey HDS post hoc test of the number of CAZymes in the Polysaccharide Lyase (PL) CAZy class across soft and hard plant tissue targeting genera in *Pectobacteriaceae* and *Musicola*, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	**Group 2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Hard	Soft	11.5549	0	10.8223	12.2874	True
Hard	Unknown	7.2767	0	4.1971	10.3562	True
Soft	Unknown	-4.2782	0.0026	-7.2882	-1.2683	True

SI Table 17: Turkey HSD test of Polysaccharide Lyase (PL) CAZy class frequencies

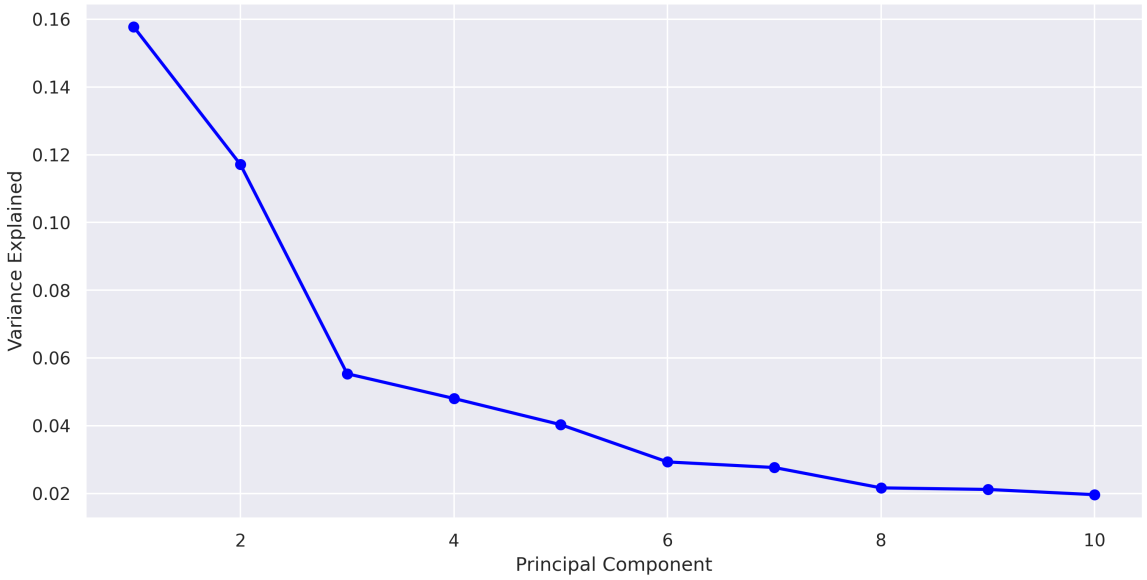
Output from a Turkey HDS post hoc test of the number of CAZymes in the Polysaccharide Lyase (PL) CAZy class across the *Pectobacteriaceae* genera, performed using the Python StatsModels package (see Methods). A p-value (P-Adj, adjusted p-value) of 0.05 was used as a cut off rejecting the null hypothesis (NH). In addition the mean difference (MeanDiff) and lower and upper bounds of the confidence interval for the mean difference.

Group 1	**Group 2**	**MeanDiff**	**P-Adj**	**Lower**	**Upper**	**Reject NH**
Acerihabitans	Affinibrenneria	0	1	-10.5548	10.5548	False
Acerihabitans	Brenneria	3.2424	0.8984	-4.3332	10.8181	False
Acerihabitans	Dickeya	15.6019	0	8.1204	23.0834	True
Acerihabitans	Lonsdalea	2.7949	0.9515	-4.7636	10.3534	False
Acerihabitans	Musicola	10.25	0.005	1.9057	18.5943	True
Acerihabitans	Pectobacterium	14.0162	0	6.5442	21.4882	True
Acerihabitans	Samsonia	7	0.4716	-3.5548	17.5548	False
Affinibrenneria	Brenneria	3.2424	0.8984	-4.3332	10.8181	False
Affinibrenneria	Dickeya	15.6019	0	8.1204	23.0834	True
Affinibrenneria	Lonsdalea	2.7949	0.9515	-4.7636	10.3534	False
Affinibrenneria	Musicola	10.25	0.005	1.9057	18.5943	True
Affinibrenneria	Pectobacterium	14.0162	0	6.5442	21.4882	True
Affinibrenneria	Samsonia	7	0.4716	-3.5548	17.5548	False
Brenneria	Dickeya	12.3595	0	10.9601	13.7589	True
Brenneria	Lonsdalea	-0.4476	0.9945	-2.2128	1.3177	False
Brenneria	Musicola	7.0076	0	3.0562	10.959	True
Brenneria	Pectobacterium	10.7738	0	9.4259	12.1217	True
Brenneria	Samsonia	3.7576	0.8033	-3.8181	11.3332	False
Dickeya	Lonsdalea	-12.8071	0	-14.1104	-11.5037	True
Dickeya	Musicola	-5.3519	0.0005	-9.1197	-1.5842	True
Dickeya	Pectobacterium	-1.5857	0	-2.2177	-0.9538	True
Dickeya	Samsonia	-8.6019	0.0118	-16.0834	-1.1204	True
Lonsdalea	Musicola	7.4551	0	3.5367	11.3735	True
Lonsdalea	Pectobacterium	11.2213	0	9.9735	12.4692	True
Lonsdalea	Samsonia	4.2051	0.693	-3.3534	11.7636	False
Musicola	Pectobacterium	3.7662	0.048	0.0173	7.5151	True
Musicola	Samsonia	-3.25	0.9364	-11.5943	5.0943	False
Pectobacterium	Samsonia	-7.0162	0.0837	-14.4882	0.4558	False

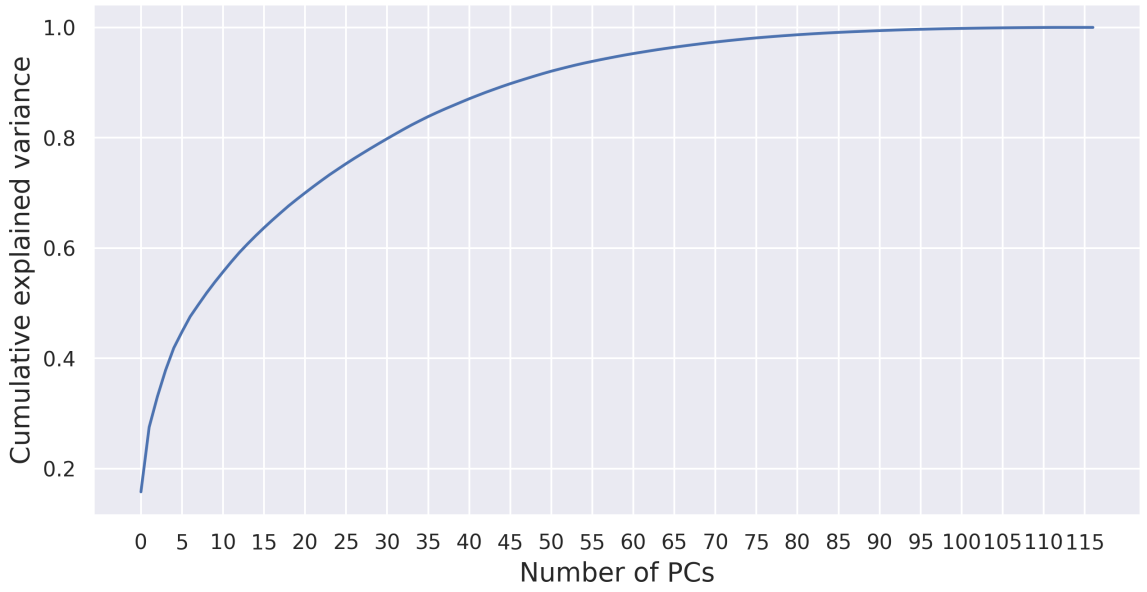
7 Principal Component Analysis (PCA) of CAZy family frequencies

SI Figure 9: Variance captured by principal component analysis of *Pectobacteriaceae* CAZy family frequencies

The number of unique protein accessions was calculated for each CAZy family, per genome. Dimensional reduction was performed on these CAZy family frequencies. This figure plots the cumulative frequency across the computed principal components (PCs). The scree plot of this figure plots the proportion of variance in the original data set that is captured by each PC. The *Pectobacteriaceae* genomes were projected onto principal components (PCs) PC1 and PC2, and *Dickeya* and *Pectobacterium* genomes located in the positive X-axis are annotated in the figure.



(a) Scree plot



(b) Cumulative explained variance

Figure 9: Principal component analysis (PCA) of the CAZy family frequencies in *Pectobacteriaceae* genomes, plotting [A] the cumulative frequency across all computed principal components (PCs). [B] Scree plot plotting the fraction of variance captured by each PC.

SI Figure 10: Principal component analysis of *Pectobacteriaceae* CAZy family frequencies PC1 and PC2

The number of unique protein accessions was calculated for each CAZy family, per genome. Dimensional reduction was performed on these CAZy family frequencies. The *Pectobacteriaceae* genomes were projected onto principal components (PCs) PC1 and PC2, annotating *Dickeya* genomes that cluster with *Musicola* genomes (figure 10).

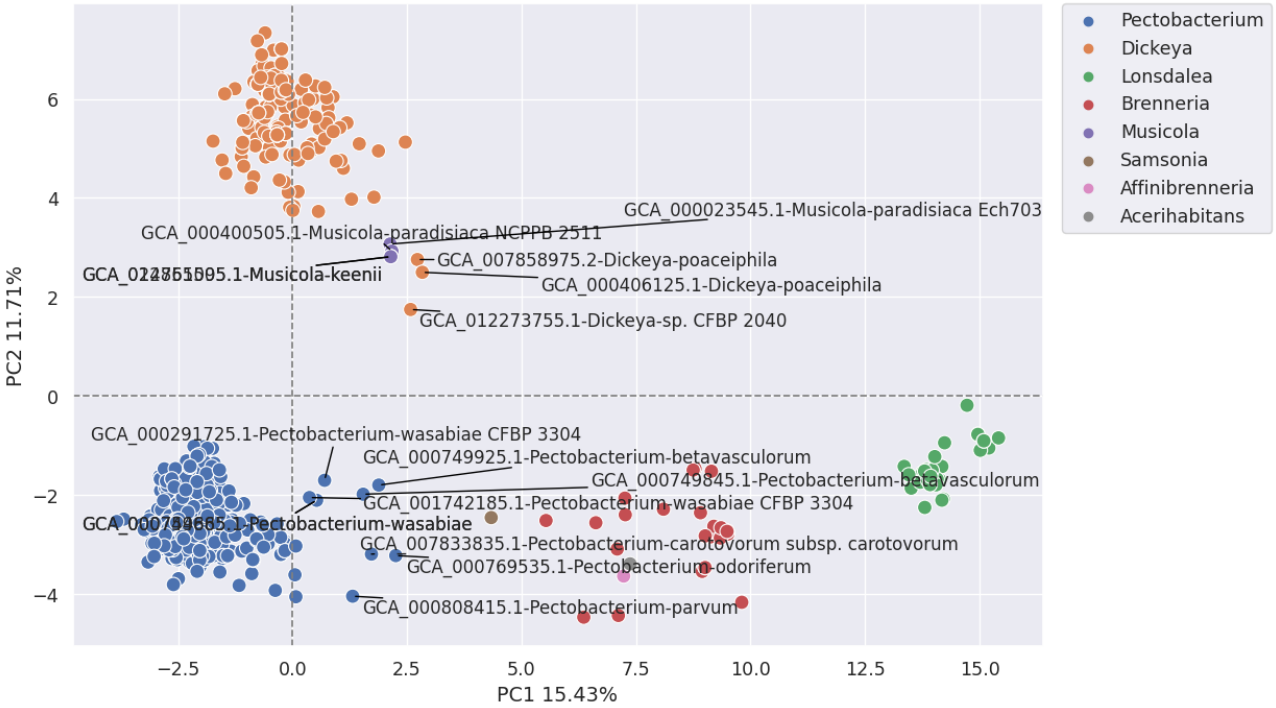


Figure 10: Principal component analysis (PCA) of the CAZy family frequencies in *Pectobacteriaceae* genomes, projecting genomes onto principal components (PCs) PC1 and PC2, colour coding genomes by genus classification. *Dickeya* genomes that cluster with *Musicola* genomes are annotated, as well as *Pectobacterium* genomes that are placed in the positive PC1-axis.

SI Table 18: Phenotypic comparison of *Musicola* and *Dickeya poaceiphila*

Carbon sources highlighted in green indicate conditions where both *D. poaceiphila* display a different phentype to both *Musicola* genomes. Carbon sources/conditions highlighted in yellow indicate conditions where one of genomes shows a different phenotype to the other three.

Carbon source	Strains	3937	A6065T (CFBP 4178T)	A3967T (CFBP 8732T)	NCPPB 569T	CFBP 2040	3937
Both genomes differs from the other 4	**Species**	*D. dadantii*	***M. paradisiaca***	***M. keenii***	***D.poaceiphila***	***D.poaceiphila***	*D.dadantii*
1 Musicola genome differ from *D. popoaceiphilaac*	**Reference**	Hugouvieux-Cotte-Pattat et al., 2021; https://doi.org/10.1099/ijsem.0.005037			Hugouvieux-Cotte-Pattat et al., 2020; https://doi.org/10.1099/ijsem.0.004306		
L-Arabinose		+	-	+	+	+	+
N-Acetyl-D-glucosamine		+	+	+	+	+	+
D-Saccharic acid (glucaric acid)		+	-	+	+	+	+
Succinic acid		w	w	+	+	+	w
D-Galactose		+	-	+	w	+	+
L-Aspartic acid		+	w	+	+	+	+
L-Proline		-	-	-	-	-	-
D-Alanine		-	-	-	-	-	-
D-Trehalose		-	-	-	-	-	-
D-Mannose		+	+	+	+	+	+
Dulcitol		-	-	-	-	-	-
D-Serine		-	-	-	-	-	-
D-Sorbitol		-	-	-	-	-	-
Glycerol		+	w	+	w	+	+
L-Fucose		-	-	-	-	-	-
D-Glucuronic acid		-	w	+	-	w	-
D-Gluconic acid		+	+	+	-	-	+
D,L-α-Glycerol-phosphate		w	w	w	w	w	w
D-Xylose		+	-	+	+	+	+
L-Lactic acid		w	-	-	+	+	w
Formic acid		w	-	-	+	+	w
D-Mannitol		+	-	-	+	+	+
L-Glutamic acid		w	-	-	w	w	w
D-Glucose-6-phosphate		+	-	+	-	-	+
D-Galactonic acid-γ-lactone		-	-	-	-	-	-
D,L-Malic acid		+	w	+	+	+	+
D-Ribose		+	-	+	+	+	w
Tween 20		-	-	-	-	-	-
L-Rhamnose		-	-	-	+	+	-
D-Fructose		+	+	+	+	+	+
Acetic acid		w	-	-	w	w	w
α-D-Glucose		+	+	+	+	+	+
Maltose		-	-	-	-	-	-
D-Melibiose		+	+	-	+	+	+
Thymidine		-	-	-	w	w	-
L-Asparagine		+	w	+	+	+	+
D-Aspartic acid		+	-	-	w	+	+
D-Glucosaminic acid		-	-	-	-	-	-
1,2-Propanediol		-	-	-	-	-	-
Tween 40		-	-	-	-	-	-
α-Keto-glutaric acid		-	-	-	-	-	-
α-Keto-butyric acid		-	-	-	-	-	-
α-Methyl-D-galactoside		+	-	-	w	w	+
α-D-Lactose		-	-	-	w	+	-
Lactulose		-	-	-	-	-	-

Table continued

Sucrose	+	+	+	+	+	+
Uridine	-	-	-	-	-	-
L-Glutamine	w	-	-	w	w	w
m-Tartaric Acid	w	-	-	+	-	w
D-Glucose-1-Phosphate	+	-	-	-	-	+
D-Fructose-6-Phosphate	+	-	+	-	-	+
Tween 80	-	-	-	-	-	-
α-Hydroxy glutaric acid-γ-lactone	-	-	-	-	-	-
α-Hydroxy butyric acid	-	-	-	-	-	-
β-Methyl-D-glucoside	+	w	+	+	+	+
Adonitol	-	-	-	-	-	-
Maltotriose	-	-	-	-	-	-
2-Deoxy adenosine	-	-	-	-	-	-
Adenosine	-	-	-	-	-	-
Glycyl-L-aspartic acid	-	-	-	-	-	-
Citric acid	+	-	-	+	+	+
myo-Inositol	+	-	+	-	-	+
D-Threonine	-	-	-	-	-	-
Fumaric acid	+	+	+	+	+	+
Bromo succinic acid	w	w	w	w	w	w
Propionic acid	-	-	-	-	-	-
Mucic acid (galactaric acid)	+	-	w	+	w	+
Glycolic acid	-	-	-	-	-	-
Glyoxylic acid	-	-	-	-	-	-
D-Cellobiose	+	-	-	-	-	+
Inosine	-	-	-	-	-	-
Glycyl-L-glutamic acid	-	-	-	-	-	-
Tricarballylic acid	-	-	-	-	-	-
L-Serine	+	-	w	+	+	+
L-Threonine	-	-	-	-	-	-
L-Alanine	-	-	w	-	w	-
L-Alanyl-glycine	-	-	-	-	-	-
Acetoacetic acid	-	-	-	-	-	-
N-Acetyl-β-D-mannosamine	-	-	-	-	-	-
Mono methyl succinate	-	-	-	-	-	-
Methyl pyruvate	+	+	+	+	w	+
D-Malic acid	w	-	w	w	w	w
L-Malic acid	+	w	+	+	+	+
Glycyl-L-proline	-	-	-	-	-	-
p-Hydroxy phenyl acetic acid	-	-	-	-	-	-
m-Hydroxy phenyl acetic acid	-	-	-	-	-	-
Tyramine	-	-	-	-	-	-
D-Psicose	-	-	+	+	+	-
L-Lyxose	-	-	w	w	w	-
Glucuronamide	-	-	-	-	-	-
Pyruvic acid	+	-	+	+	+	+
L-Galactonic acid-γ-lactone	-	-	-	-	-	-
D-Galacturonic acid	+	+	+	+	+	+

Table continued

Phenylethyl-amine	-	-	-	-	-	-
2-Aminoethanol	-	-	-	-	-	-
Chondroitin sulfate C	-	-	-	-	-	-
α-Cyclodextrin	-	-	-	-	-	-
β-Cyclodextrin	-	-	-	-	-	-
γ-Cyclodextrin	-	-	-	-	-	-
Dextrin	-	-	-	-	-	-
Gelatin	-	-	-	-	-	-
Glycogen	-	-	-	-	-	-
Inulin	-	-	-	-	-	-
Laminarin	-	-	-	-	-	-
Mannan	-	-	-	-	-	-
Pectin	+	+	+	-	-	+
N-Acetyl-D-galactosamine	-	-	-	-	-	-
N-Acetyl-neuraminic acid	-	-	-	-	-	-
β-D-Allose	-	-	-	-	-	-
Amygdalin	-	-	-	-	-	-
D-Arabinose	w	-	-	+	+	w
D-Arabitol	-	-	-	-	-	-
L-Arabitol	-	-	-	-	-	-
Arbutin	+	+	+	+	+	+
2-Deoxy-D-ribose	-	-	-	-	-	-
I-Erythritol	-	-	-	-	-	-
D-Fucose	-	-	-	-	-	-
3-0-β-D-Galacto-pyranosyl-D-arabinose	-	-	-	-	-	-
Gentiobiose	-	-	-	+	+	-
L-Glucose	-	-	-	-	-	-
Lactitol	-	-	-	-	-	-
D-Melezitose	-	-	-	-	-	-
Maltitol	-	-	-	-	-	-
a-Methyl-D-glucoside	-	-	-	-	-	-
β-Methyl-D-galactoside	w	-	-	w	w	w
3-Methyl Glucose	-	-	-	-	-	-
β-Methyl-D-glucuronic acid	-	-	-	-	-	-
α-Methyl-D-mannoside	-	-	-	-	-	-
β-Methyl-D-xyloside	-	-	-	-	-	-
Palatinose	-	-	-	-	-	-
D-Raffinose	+	w	-	+	+	+
Salicin	+	+	+	+	+	+
Sedoheptulosan	-	-	-	-	-	-
L-Sorbose	-	-	-	-	-	-
Stachyose	-	-	-	-	-	-
D-Tagatose	-	-	-	w	w	-
Turanose	-	-	-	-	-	-
Xylitol	-	-	-	-	-	-
N-Acetyl-D-glucosaminitol	-	-	-	-	-	-
γ-Amino butyric acid	-	-	-	-	-	-
δ-Amino valeric acid	-	-	-	-	-	-
Butyric acid	-	-	-	-	-	-

Table continued

Capric acid	-	-	-	-	-	-
Caproic acid	-	-	-	-	-	-
Citraconic acid	-	-	-	-	-	-
Citramalic acid	-	-	-	-	-	-
D-Glucosamine	+	+	+	w	w	+
2-Hydroxy benzoic acid	-	-	-	-	-	-
4-Hydroxy benzoic acid	-	-	-	-	-	-
β-Hydroxy butyric acid	-	-	-	-	-	-
γ-Hydroxy butyric acid	-	-	-	-	-	-
a-Keto-valeric acid	-	-	-	-	-	-
Itaconic acid	-	-	-	-	-	-
5-Keto-D-gluconic acid	-	-	-	-	-	-
D-Lactic acid methyl ester	-	-	-	-	-	-
Malonic acid	w	-	-	w	w	w
Melibionic acid	-	-	-	-	-	-
Oxalic acid	-	-	-	-	-	-
Oxalomalic acid	-	-	-	-	-	-
Quinic acid	-	-	-	-	-	-
D-Ribono-1,4-lactone	-	-	-	-	-	-
Sebacic acid	-	-	-	-	-	-
Sorbic acid	-	-	-	-	-	-
Succinamic acid	+	w	+	+	+	+
D-Tartaric acid	-	-	-	-	-	-
L-Tartaric acid	-	-	-	-	-	-
Acetamide	-	-	-	-	-	-
L-Alaninamide	-	-	-	-	-	-
N-Acetyl-L-glutamic acid	-	-	-	-	-	-
L-Arginine	-	-	-	-	-	-
Glycine	-	-	-	-	-	-
L-Histidine	-	-	-	-	-	-
L-Homoserine	-	-	-	-	-	-
Hydroxy-L-proline	-	-	-	-	-	-
L-Isoleucine	-	-	-	-	-	-
L-Leucine	-	-	-	-	-	-
L-Lysine	-	-	-	-	-	-
L-Methionine	-	-	-	-	-	-
L-Ornithine	-	-	-	-	-	-
L-Phenylalanine	-	-	-	-	-	-
L-Pyroglutamic acid	-	-	-	-	-	-
L-Valine	-	-	-	-	-	-
D,L-Carnitine	-	-	-	-	-	-
Sec-Butylamine	-	-	-	-	-	-
D.L-Octopamine	-	-	-	-	-	-
Putrescine	-	-	-	-	-	-
Dihydroxy Acetone	-	-	-	-	-	-
2,3-Butanediol	-	-	-	-	-	-
2,3-Butanedione	-	-	-	-	-	-
3-Hydroxy 2-Butanone	-	-	-	-	-	-

SI Table 19: Genomes with potentially less rich CAZomes

Table listing the assembly status and CheckM analysis of the genomic sequence completeness as reported in the NCBI Assembly database (October 2023) for genomes that contained fewer families than genomes from the same genus (as identified in SI figures 10-13).

Assembly status and CheckM genomic assembly analysis reported in the NCBI Assembly database (October 2023)

Assembly Accession	**Genus**	**Species**	**Assembly status**	**CheckM analysis (% (percentile)**
GCA_029023745.1	*Pectobacterium*	*colocasium* PL155	Complete	NA
GCA_021907015.1	*Pectobacterium*	sp. PL152	Complete	NA
GCA_000808415.1	*Pectobacterium*	*parvum* Y1	Scaffold	81.03 (13th)
GCA_004137815.1	*Pectobacterium*	*zantedeschiae* 2M	Scaffold	89.31 (33rd)
GCA_012273755.1	*Dickeya*	sp. CFBP 2040	Contig	90.95 (100th)
GCA_000406125.1	*Dickeya*	*poaceiphila* NCPPB 569	Chromosome	89.80 (50th)
GCA_013168485.1	*Dickeya*	*dadantii* Aka1-1	Contig	83.55 (11th)
GCA_007858975.2	*Dickeya*	*poaceiphila* NCPPB 569	Complete	93.97 (100th)
GCA_002111555.1	*Lonsdalea*	*populi* CFCC11748	Contig	91.09 (100th)
GCA_003269855.1	*Lonsdalea*	*populi* CFCC11200	Contig	100 (100th)
GCA_003666245.1	*Brenneria*	*alni* NCPPB 3934	Contig	100 (100th)
GCA_000406125.1	*Brenneria*	*rubrifaciens* 6D370	Complete	100 (100th)

8 Exploration of CAZy family frequencies

SI Table 20: Genus specific families in *Pectobacteriaceae*

CAZy families that are only found in one genus of *Pectobacteriaceae*, or only found in soft or hard plant tissue targeting genomes are listed in table 1.

Table 1: CAZy families found in only one genus, or only soft plant tissue targeting genomes, or hard plant tissue targeting genomes in *Pectobacteriacea*

Genus	**GH**	**GT**	**CE**	**PL**	**AA**	**CBM**
Pectobacterium	GH121, GH146, GH18	GT101, GT102, GT11, GT111, GT14, GT24, GT52		PL11	AA10	CBM13
Dickeya	GH113, GH148, GH25, GH91	GT97	CE2	PL10		CBM4
Soft tissue	GH16, GH18, GH25, GH91, GH113, GH121, GH146, GH148	GT11, GT14, GT24, GT25, GT52, GT97, GT101, GT102, GT111	CE2	PL10, PL11, PL35	AA10	CBM0, CBM4, CBM13, CBM91
Brenneria	GH106	GT21		PL17		
Acerihabitans	GH127, GH15					
Hard tissue	GH15, GH37, GH39, GH51, GH106, GH127, GH140	GT20, GT21, GT28		PL17		

SI Figure 11: Core CAZome Families

The number of unique protein accessions was calculated for each CAZy family, per genome, and the mean number of proteins per CAZy family that was found in every genome (the "core CAZome") was plotted as a proportional area plot, breaking down the data per genus.

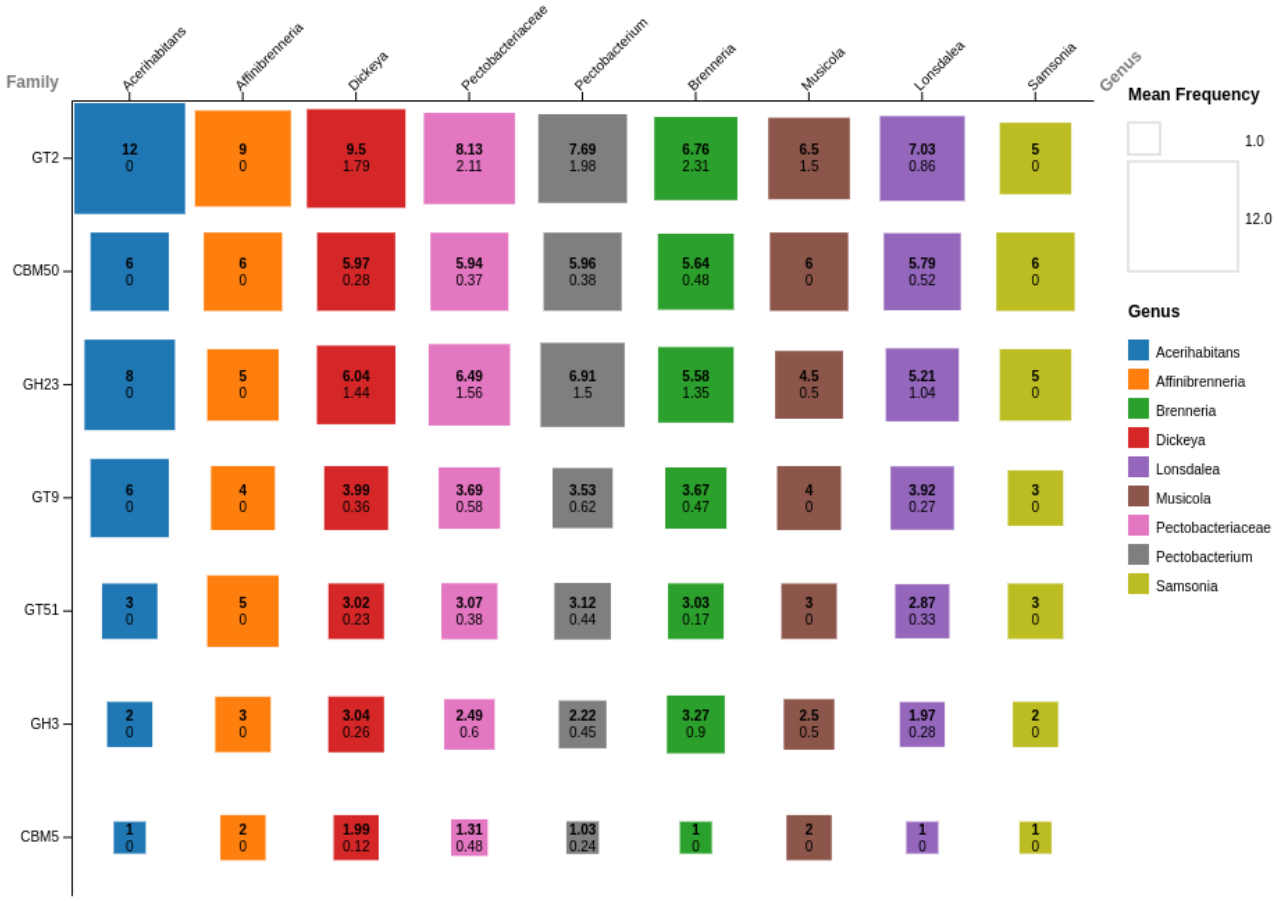
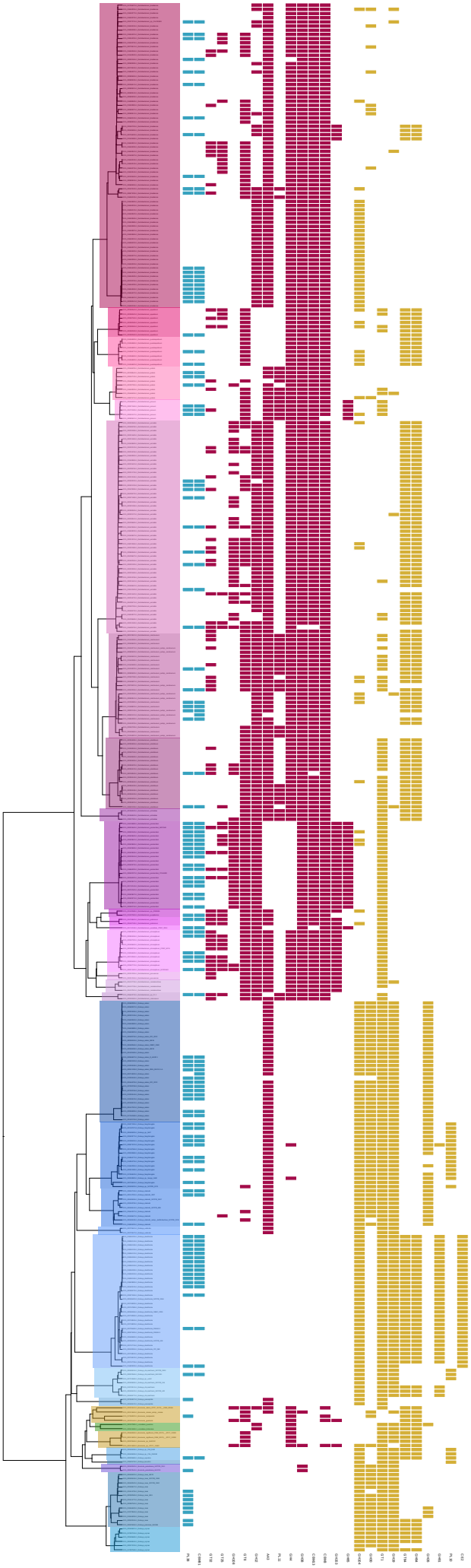


Figure 11: The core CAZome of *Pectobacteriaceae*. Proportional area plot of the mean number of CAZymes per CAZyme family that are present in all *Pectobacteriaceae* genomes (i.e. the core CAZome). Each square is annotated with the mean frequency (in bold), and standard deviation.

9 Coinfinder output

SI Figure 12: Output from `coinfinder` with taxonomic annotations included (on the next page)


PLM
CBM1
GT2
GT4
PHOS
GT1
CHI
CAT
TLL
PEP
GH5
CBM3
CHS1
GH3
CHS2
GH9
GT3
GH6
GH5
PLM
CEC

SI Figure 13: ANI versus CAZyme family frequency tanglegram

The number of unique protein accessions was calculated for each CAZy family, per genome. These CAZy family frequencies were used to build a dendrogram. An ANI-based tree was built using all genomic sequences and `pyani`.

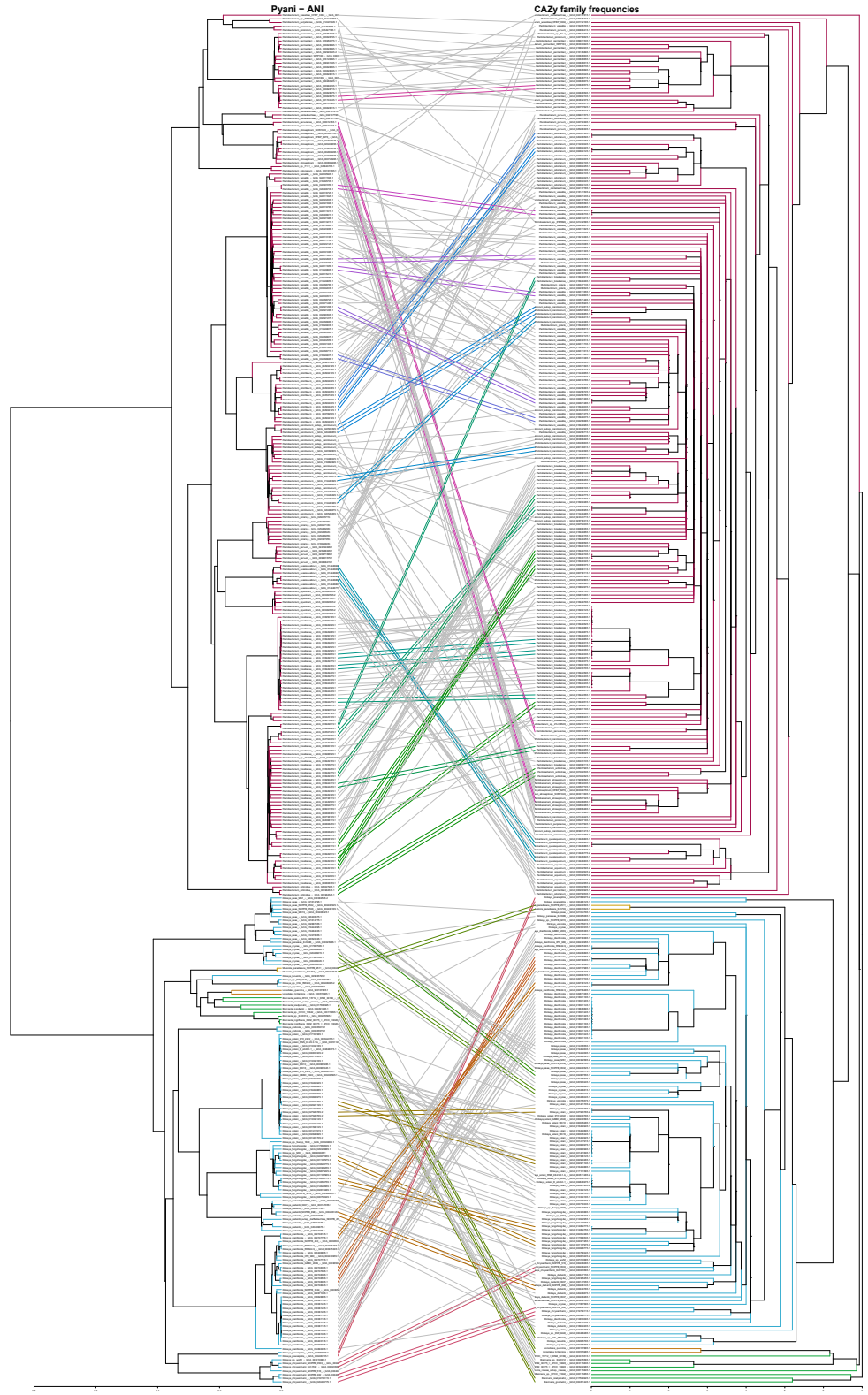


Figure 12: Tanglegram comparing the topologies of dendrograms generated from single linkage hierarchical clustering of CAZyme family frequencies and average nucleotide identities (ANI). *Leaf branches are colour coded by the genus classification of the corresponding genomes: pink, Pectobacterium; green, Brenneria; purple, Affinibrenneria; teal, Acerihabitans; gold, Musicola; blue, Dickeya; orange, Lonsdalea. Matching subtrees are indicated by colour-coded joining lines, matching the corresponding genomes between the dendrograms.*